

Homework 4 Submission

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Part A

Throughout this part, all logarithms are natural and universal constants may change from line to line.

1. **VC dimension through supports.** For $I \subseteq \{1, \dots, d\}$ with $|I| = k$, let $\mathcal{H}_I$ be the class of homogeneous halfspaces whose weight vector is supported on I . Then

$$\mathcal{H}_k = \bigcup_{I \subseteq [d], |I|=k} \mathcal{H}_I,$$

because a vector with fewer than k nonzero coordinates is supported on many k -subsets. For each fixed I , $\mathcal{H}_I$ is just the class of homogeneous halfspaces in $\mathbb{R}^k$, so

$$\text{VCdim}(\mathcal{H}_I) \leq k.$$

We bound $\text{VCdim}(\mathcal{H}_k)$ by showing that any set of m points shattered by $\mathcal{H}_k$ must satisfy $m = O(k \log(e \frac{d}{k}))$. Since $\text{VCdim}(\mathcal{H}_k)$ is the largest shattered size, this is exactly the desired bound. So fix any candidate set of m points. If $m < k$, the bound $m = O(k \log(e \frac{d}{k}))$ is immediate, because $\log(e \frac{d}{k}) \geq 1$ gives $m < k \leq k \log(e \frac{d}{k})$. Thus assume $m \geq k$. Then, by Sauer–Shelah, on these m points each fixed-support class realizes at most

$$\Gamma_{\mathcal{H}_I}(m) \leq \sum_{j=0}^k \binom{m}{j} \leq \left(e \frac{m}{k}\right)^k.$$

The union over supports can realize no more patterns than the sum of the patterns realized by the fixed-support classes. Therefore

$$\Gamma_{\mathcal{H}_k}(m) \leq \binom{d}{k} \left(e \frac{m}{k}\right)^k \leq \left(e \frac{d}{k}\right)^k \left(e \frac{m}{k}\right)^k.$$

If $\mathcal{H}_k$ shatters m points, then

$$2^m \leq \left(e^2 d \frac{m}{k^2}\right)^k,$$

i.e. $m \log(2) \leq k \log(e^2 \frac{d}{k}) + k \log(\frac{m}{k})$. Setting $u := \frac{m}{k}$ and $A := e^2 \frac{d}{k}$ this is

$$u \log(2) \leq \log(A) + \log(u).$$

Since $\log(u) \leq \left(\frac{\log(2)}{2}\right)u$ for all $u \geq 4$, either $u < 4$ or $\left(\frac{\log(2)}{2}\right)u \leq \log(A)$, so

$$u \leq \max\left\{4, \left(\frac{2}{\log(2)}\right) \log(A)\right\} = O\left(\log\left(e \frac{d}{k}\right)\right).$$

Hence $m = ku = O(k \log(e \frac{d}{k}))$, and therefore

$$\text{VCdim}(\mathcal{H}_k) = O\left(k \log\left(e \frac{d}{k}\right)\right).$$

2. **Penalty-based SRM.** Choose any prior $p_k > 0$ with $\sum_{k=1}^d p_k \leq 1$; for example,

$$p_k = \frac{6}{\pi^2 k^2}.$$

Let

$$\text{pen}(k, n, \delta) := C_0 \sqrt{\frac{k \log\left(e \frac{d}{k}\right) + \log\left(\frac{1}{p_k}\right) + \log\left(\frac{1}{\delta}\right)}{n}}$$

for a large enough universal constant C_0 , and define

$$\hat{h} \in \arg \min_{1 \leq k \leq d, h \in \mathcal{H}_k} \{L_{S(h)} + \text{pen}(k, n, \delta)\}.$$

Proof. By item 1 and the class-level SRM theorem, with probability at least $1 - \delta$, simultaneously for every k and every $h \in \mathcal{H}_k$,

$$|L_{\mathcal{D}}(h) - L_{S(h)}| \leq C_1 \sqrt{\frac{\text{VCdim}(\mathcal{H}_k) + \log\left(\frac{1}{p_k}\right) + \log\left(\frac{1}{\delta}\right)}{n}} \leq \text{pen}(k, n, \delta)$$

after increasing C_0 .

On this event, let $\hat{k}$ be an index used by the SRM rule together with $\hat{h}$. For any k and $h \in \mathcal{H}_k$,

$$L_{\mathcal{D}}(\hat{h}) \leq L_{S(\hat{h})} + \text{pen}(\hat{k}, n, \delta) \leq L_{S(h)} + \text{pen}(k, n, \delta) \leq L_{\mathcal{D}}(h) + 2 \text{pen}(k, n, \delta).$$

Taking the infimum over k, h gives

$$L_{\mathcal{D}}(\hat{h}) \leq \inf_{1 \leq k \leq d, h \in \mathcal{H}_k} \left\{ L_{\mathcal{D}}(h) + C \sqrt{\frac{k \log\left(e \frac{d}{k}\right) + \log\left(\frac{1}{p_k}\right) + \log\left(\frac{1}{\delta}\right)}{n}} \right\}.$$

□

The term $k \log\left(e \frac{d}{k}\right)$ is the statistical cost of searching over sparse predictors with a fixed sparsity level: it contains the cost of choosing one of about $\binom{d}{k}$ supports and the cost of fitting a k -dimensional halfspace on that support. The term $\log\left(\frac{1}{p_k}\right)$ is the additional price for allowing the data to choose the sparsity level.

3. **Comparison with dense halfspaces.** Suppose w^* is s -sparse and $L_{\mathcal{D}}(w^*) \leq \eta$. Applying the oracle inequality with $k = s$ and $h = w^*$ gives, with probability at least $1 - \delta$,

$$L_{\mathcal{D}}(\hat{h}) \leq \eta + C \sqrt{\frac{s \log\left(e \frac{d}{s}\right) + \log\left(\frac{1}{p_s}\right) + \log\left(\frac{1}{\delta}\right)}{n}}.$$

Thus it is enough to take

$$n \geq C' \frac{s \log\left(e \frac{d}{s}\right) + \log\left(\frac{1}{p_s}\right) + \log\left(\frac{1}{\delta}\right)}{\varepsilon^2}.$$

With the concrete prior $p_s = \frac{6}{\pi^2 s^2}$, this is

$$n = O\left(\frac{s \log(e \frac{d}{s}) + \log(s) + \log(\frac{1}{\delta})}{\varepsilon^2}\right).$$

Learning over all homogeneous halfspaces in $\mathbb{R}^d$ gives the agnostic sample size

$$O\left(\frac{d + \log(\frac{1}{\delta})}{\varepsilon^2}\right).$$

The sparse bound is smaller when

$$s \log\left(e \frac{d}{s}\right) + \log\left(\frac{1}{p_s}\right) \ll d.$$

In particular, it improves substantially in the genuinely sparse regime $s \ll d$, up to the logarithmic support-selection factor. When s is comparable to d , the sparse and dense bounds are of the same order.

4. **Validation as model selection.** Write $m = \frac{n}{2}$. For each k , let w_k be an ERM over $\mathcal{H}_k$ on S_1 , and choose $\hat{k}$ by minimizing $L_{S_2}(w_k)$. Let $\hat{w} := w_{\hat{k}}$.

Proof. First use a VC uniform-convergence bound on S_1 , with failure probability $\frac{\delta}{2d}$ for each k . With probability at least $1 - \frac{\delta}{2}$, for every k and every $w \in \mathcal{H}_k$,

$$|L_{\mathcal{D}}(w) - L_{S_1}(w)| \leq \alpha_k := C_1 \sqrt{\frac{k \log(e \frac{d}{k}) + \log(\frac{d}{\delta})}{n}},$$

where the factor $m = \frac{n}{2}$ has been absorbed into the constant. On this event, ERM optimality gives, for every $w \in \mathcal{H}_k$,

$$L_{\mathcal{D}}(w_k) \leq L_{S_1}(w_k) + \alpha_k \leq L_{S_1}(w) + \alpha_k \leq L_{\mathcal{D}}(w) + 2\alpha_k.$$

Now condition on S_1 . The candidates $w_1, \dots, w_d$ are fixed, so the validation step is a finite-class selection problem of size at most d . By Hoeffding and a union bound, with conditional probability at least $1 - \frac{\delta}{2}$,

$$|L_{\mathcal{D}}(w_k) - L_{S_2}(w_k)| \leq \beta := C_2 \sqrt{\frac{\log(\frac{d}{\delta})}{n}}$$

for all k . On this event, for every k ,

$$L_{\mathcal{D}}(\hat{w}) \leq L_{S_2}(\hat{w}) + \beta \leq L_{S_2}(w_k) + \beta \leq L_{\mathcal{D}}(w_k) + 2\beta.$$

Combining the two displays and using $\beta \leq \alpha_k$ gives

$$L_{\mathcal{D}}(\hat{w}) \leq L_{\mathcal{D}}(w) + C \sqrt{\frac{k \log(e \frac{d}{k}) + \log(\frac{d}{\delta})}{n}}$$

for every k and every $w \in \mathcal{H}_k$. Taking the infimum proves

$$L_{\mathcal{D}}(\hat{w}) \leq \inf_{1 \leq k \leq d, w: |\text{supp}(w)| \leq k} \left\{ L_{\mathcal{D}}(w) + C \sqrt{\frac{k \log(e \frac{d}{k}) + \log(\frac{d}{\delta})}{n}} \right\}.$$

□

The validation rule uses S_1 only to fit one candidate predictor at each sparsity level, and it uses S_2 only to choose among the d fitted candidates. Thus the model-selection price on the validation half is just $\log(d)$. The price is that fitting uses only half the sample, and the final choice cannot revisit all predictors at a level k after looking at S_2 . The penalty-based SRM rule uses the full sample for both fitting and choosing k , but it must include an explicit prior cost $\log\left(\frac{1}{p_k}\right)$ for the complexity level.

5. **Computation.** In the realizable case, the direct brute-force algorithm is:

- enumerate all supports $I \subseteq [d]$ with $|I| = k$;
- for each I , solve the feasibility problem in variables $w_I \in \mathbb{R}^k$:

$$w_I^T x_{i,I} \geq 0 \quad \text{for all } i \text{ with } y_i = +1,$$

and

$$w_I^T x_{i,I} \leq -1 \quad \text{for all } i \text{ with } y_i = -1;$$

- if one support is feasible, output the corresponding k -sparse vector, with all coordinates outside I set to 0.

The normalization ≤ -1 for negative examples is only a way to replace strict inequalities by closed linear constraints. If a support admits a consistent classifier and there is at least one negative example, then every negative score is strictly below 0 under the convention $\text{sign}(0) = +1$; because the sample is finite, scaling w makes all negative scores at most -1 while preserving the nonnegative scores on positive examples. If there are no negative examples, the zero vector already labels all sample points by $+1$ under the convention $\text{sign}(0) = +1$.

There are

$$\binom{d}{k} \leq \left(e \frac{d}{k}\right)^k$$

supports. Linear feasibility in k variables and n constraints is solvable in time polynomial in n and k (ignoring standard bit-complexity bookkeeping). Hence the runtime is

$$\binom{d}{k} \cdot \text{poly}(n, k) \leq \left(e \frac{d}{k}\right)^k \cdot \text{poly}(n, k).$$

This is polynomial in d for constant k . For $k = O(\log(d))$ it is already quasi-polynomial in general, and for k growing as a power of d or as $\Theta(d)$ it is super-polynomial or exponential.

Thus the sample complexity and the computation tell different stories. Statistically, k -sparse halfspaces have complexity only about $k \log(e \frac{d}{k})$. Computationally, the naive search over supports pays the much larger combinatorial factor $\binom{d}{k}$. This is exactly the Week 4 distinction: a class can be sample-efficient because it has small VC complexity, while ERM or consistency search over that class can still be computationally expensive.

Part B

1. **VC dimension and point-posterior PAC-Bayes.** The threshold class has VC dimension 1.

Proof. A singleton $\{x\}$ is shattered: choose $t = x$ to label it by 1, and choose $t = x + 1$ to label it by 0. On the other hand, no two points $x_1 < x_2$ can be shattered, because the labeling

$$h(x_1) = 1, \quad h(x_2) = 0$$

is impossible. Indeed, if $h_{t(x_1)} = 1$, then $t \leq x_1 < x_2$, so $h_{t(x_2)} = 1$ as well. Hence $\text{VCdim}(\mathcal{H}_N) = 1$. $\square$

Let P be uniform on the $N + 1$ thresholds. If $Q_S = \delta_{h_i}$ is a point mass on the deterministic ERM output, then

$$\text{KL}(Q_S \parallel P) = \log\left(\frac{1}{P(h_i)}\right) = \log(N + 1).$$

In the realizable setting $L_{S(Q_S)} = 0$, so the stated PAC-Bayes theorem gives, with probability at least $1 - \delta$,

$$L_{\mathcal{D}}(Q_S) \leq \sqrt{\frac{\log(N + 1) + \log\left(2\frac{n}{\delta}\right)}{2(n - 1)}}.$$

This certificate pays $\log(N + 1)$ because a point posterior has to identify one exact threshold under the uniform prior. VC theory does not pay this price for thresholds because it uses the order structure of the class: thresholds have only VC dimension 1, no matter how large the finite ordered domain is.

2. **Version-space posterior.** Let

$$a(S) = \max\{x_i : y_i = 0\}$$

with $a(S) = 0$ if there are no negative examples, and let

$$b(S) = \min\{x_i : y_i = 1\}$$

with $b(S) = N + 1$ if there are no positive examples.

Proof. A threshold h_t is consistent with a negative example $(x_i, 0)$ iff $h_{t(x_i)} = 0$, which means $x_i < t$. Thus consistency with all negative examples is equivalent to $a(S) < t$. Similarly, h_t is consistent with a positive example $(x_i, 1)$ iff $x_i \geq t$. Consistency with all positive examples is therefore equivalent to $t \leq b(S)$. Combining the two conditions,

$$V(S) = \{t : a(S) < t \leq b(S)\}.$$

Since thresholds are indexed by integers, this set has size

$$|V(S)| = b(S) - a(S).$$

$\square$

Let Q_V be uniform on the consistent thresholds. Under the uniform prior P ,

$$\text{KL}(Q_V \parallel P) = \sum_{t \in V(S)} \frac{1}{|V(S)|} \log\left(\frac{\frac{1}{|V(S)|}}{\frac{1}{N+1}}\right) = \log\left(\frac{N+1}{|V(S)|}\right).$$

Also $L_{S(Q_V)} = 0$, because every threshold in $V(S)$ is individually consistent with the sample. Compared with a point posterior, the KL term is smaller by $\log(|V(S)|)$. Thus when the version space is large, spreading the posterior across all consistent thresholds can give a much tighter PAC-Bayes certificate without increasing empirical risk.

3. **Spreading helps KL, but can hurt true risk.** Let labels be generated by h_τ and let the marginal on $\mathcal{X}_N$ be uniform. For a fixed threshold t , the two thresholds h_t and h_τ disagree exactly on the integer points between t and τ . There are $|t - \tau|$ such points, so

$$L_{\mathcal{D}}(h_t) = \frac{|t - \tau|}{N}.$$

Therefore, for the uniform posterior Q_W ,

$$L_{\mathcal{D}}(Q_W) = \frac{1}{|W|} \sum_{t \in W} L_{\mathcal{D}}(h_t) = \frac{1}{N|W|} \sum_{t \in W} |t - \tau|.$$

A concrete example is $N = 20$, $\tau = 10$, and

$$S = ((1, 0), (20, 1)).$$

This sample is realizable by h_{10} . Here $a(S) = 1$, $b(S) = 20$, and hence

$$V(S) = \{2, 3, \dots, 20\}, \quad |V(S)| = 19.$$

The uniform version-space posterior satisfies

$$\text{KL}(Q_{V(S)} \parallel P) = \log\left(\frac{21}{19}\right) < \log(21) = \text{KL}(\delta_{h_{10}} \parallel P).$$

However,

$$L_{\mathcal{D}}(Q_{V(S)}) = \frac{1}{20 \cdot 19} \sum_{t=2}^{20} |t - 10| = \frac{91}{380} > 0 = L_{\mathcal{D}}(\delta_{h_{10}}).$$

There is no contradiction with PAC-Bayes. The theorem gives an upper bound on true risk in terms of empirical risk and KL; it does not say that lower KL always means lower true risk. In this example the version-space posterior has a better complexity term, but it spreads mass over thresholds that are far from the true threshold and therefore makes mistakes on many unsampled points.

4. **Every fixed prior can be attacked.** Let P be any prior over $\mathcal{H}_N$ and assume $N \geq 3$. There are $N - 1$ internal thresholds $\tau \in \{2, \dots, N\}$. Since their total prior mass is at most 1, at least one internal threshold satisfies

$$P(h_{\tau}) \leq \frac{1}{N - 1}.$$

Fix such a τ .

Under $\mathcal{D}_{\tau}$, each sample point is either $(\tau - 1, 0)$ or $(\tau, 1)$, each with probability $\frac{1}{2}$. The probability that an n -sample fails to contain both support points is the probability that all draws are the first point or all draws are the second point:

$$2 \cdot \left(\frac{1}{2}\right)^n = 2^{1-n}.$$

Hence with probability at least $1 - 2^{1-n}$, the sample contains both support points.

On this event,

$$a(S) = \tau - 1 \quad \text{and} \quad b(S) = \tau,$$

so

$$V(S) = \{t : \tau - 1 < t \leq \tau\} = \{\tau\}.$$

Therefore any posterior Q with $L_{S(Q)} = 0$ must put all its mass on h_{τ} ; otherwise it gives positive mass to a threshold that misclassifies one of the two observed support points. Thus $Q = \delta_{h_{\tau}}$. Consequently,

$$\text{KL}(Q \parallel P) = \log\left(\frac{1}{P(h_\tau)}\right) \geq \log(N - 1),$$

with the convention that the KL is infinite if $P(h_\tau) = 0$.

This shows a limitation of zero-empirical-error PAC-Bayes certificates with a fixed prior: an adversarially chosen realizable threshold can force every consistent posterior to concentrate on a hypothesis with prior mass at most $\frac{1}{N-1}$. It does not prove that thresholds require $\log(N)$ samples to learn. VC theory gives a distribution-free learning guarantee for thresholds with VC dimension 1. The lower bound is about this particular style of fixed-prior, zero-training-error certificate, not about the information-theoretic sample complexity of threshold learning.

Part C

I significantly used AI in this assignment in order to generate the proofs. I was only able to spot-check the proofs, and I'm convinced of their correctness, however I admittedly didn't check them fully due to lack of time. I'm happy to accept deductions for partly using AI as an oracle, unfortunately I was unable to finish the homework otherwise, so I'm happy to be graded as if I had not submitted parts of it.